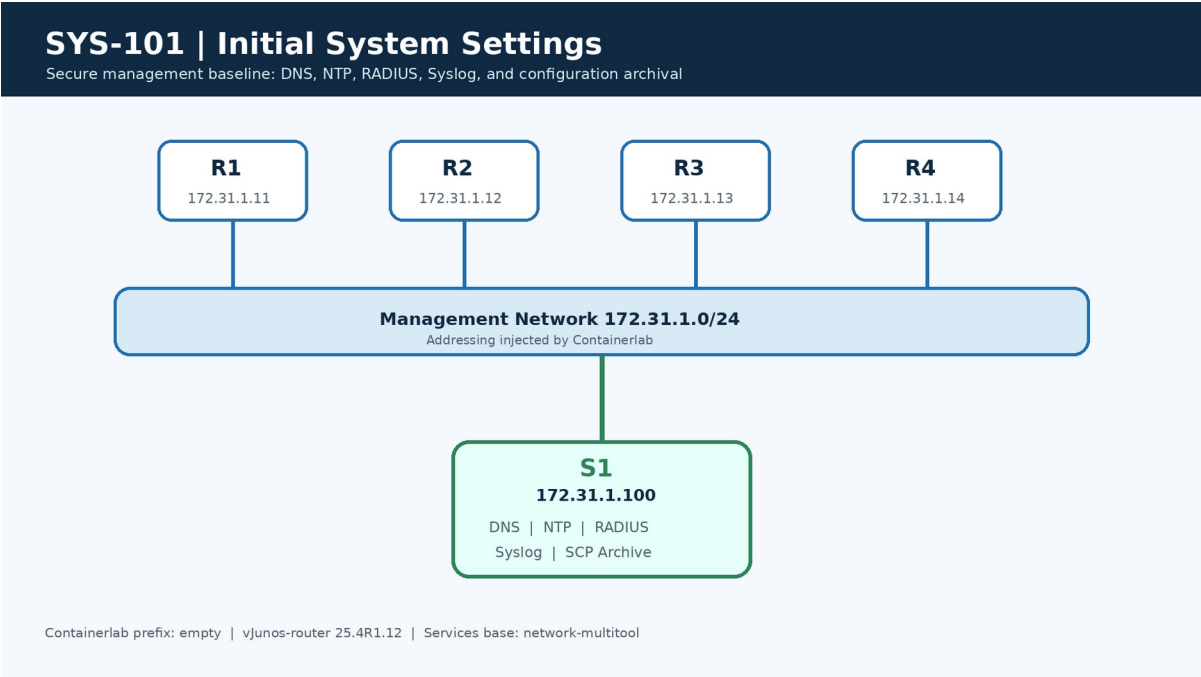


JNCIE-SP MODULAR LAB

SYS-101

Initial System Settings

Task and Solution Guide



PLATFORM	DURATION	TARGET
vJunos-router 25.4	3 hours	2 repetitions

Solution Guide — Revision 3

Platform: Containerlab, four vJunos-router 25.4R1.12 nodes, one consolidated services node

Estimated time: 3 hours | **Difficulty:** Intermediate | **Target repetitions:** 2

Lab purpose

Build and verify a secure management baseline on R1-R4. Containerlab/vrnetlab injects management addressing, **management-instance**, **mgmt_junos**, and its default route. Startup files deliberately do not configure those objects.

Addressing

Node	Containerlab management IP	Loopback	Final hostname	Role
R1	172.31.1.11/24	192.0.2.11/32	R1	Junos router
R2	172.31.1.12/24	192.0.2.12/32	R2	Junos router
R3	172.31.1.13/24	192.0.2.13/32	R3	Junos router
R4	172.31.1.14/24	192.0.2.14/32	R4	Junos router
S1	172.31.1.100/24	—	S1	DNS, NTP, RADIUS, Syslog and SCP archive

Inside each vJunos VM, **fxp0.0** is **10.0.0.15/24** and the management next hop is **10.0.0.2**. The **172.31.1.x** addresses remain the Containerlab-facing addresses used from the host and by S1.

Task 1 - Validate baseline and identity

Objective

Confirm startup configuration loading, injected management connectivity, S1 reachability, and production identity.

Requirements

1. Deploy all five nodes and verify that they are operational.
2. Confirm Containerlab management addresses and the injected **fxp0.0** configuration.
3. Change bootstrap hostnames to R1, R2, R3, and R4.
4. Save deployment and management-interface evidence.

Solution

Deploy and inspect the lab:

```
make build-services
make deploy
containerlab inspect --topo topology.clab.yml
```

On every router, replace **<NODE>** with its node name:

```
configure
set system host-name <NODE>
commit confirmed 5
run show interfaces terse fxp0
run ping 172.31.1.100 routing-instance mgmt_junos count 3
commit
```

Do not add management configuration to the startup file. An internal address such as **10.0.0.15/24** is expected on **fxp0.0**; the host still reaches the node through the Containerlab address.

Task 2 - Central and local Syslog

Objective

Create troubleshooting evidence before implementing the remaining tasks.

Requirements

5. **jncie-sp-messages** must record information and interactive commands.
6. **user-commands** must record interactive commands.
7. S1 must receive notice messages and configuration changes through **mgmt_junos**.
8. User **ops** must receive warning messages when that user is logged in.
9. Archive three local files with a size of 100 KB each.

Solution

```
configure
set system syslog archive size 100k
set system syslog archive files 3
set system syslog file jncie-sp-messages any info
set system syslog file jncie-sp-messages interactive-commands any
set system syslog file user-commands interactive-commands any
set system syslog host 172.31.1.100 any notice
set system syslog host 172.31.1.100 change-log any
set system syslog host 172.31.1.100 routing-instance mgmt_junos
set system syslog user ops any warning
commit
```

The **ops** target is valid before Task 7 creates the corresponding login template. It becomes useful whenever that identity is logged in.

Verify locally and on S1:

```
show configuration system syslog | display set
show log jncie-sp-messages | last 20
show log user-commands | last 20

find evidence/services -name '*.log' -size +0c -ls
```

Task 3 - Management services and retained route

Objective

Enable required management protocols and preserve a remote management route if RPD is unavailable.

Requirements

10. Enable SSH, Telnet, and FTP.
11. Install **10.10.10.0/24** through **10.0.0.2** in **mgmt_junos**.
12. Retain the route if RPD stops.
13. Prevent the route from being advertised by dynamic routing protocols.

Solution

```
configure
set system services ssh
set system services telnet
set system services ftp
```

```

set routing-instances mgmt_junos routing-options static route 10.10.10.0/24 next-hop 10.0.0.2
set routing-instances mgmt_junos routing-options static route 10.10.10.0/24 retain
set routing-instances mgmt_junos routing-options static route 10.10.10.0/24 no-readvertise
commit confirmed 5
run show route table mgmt_junos.inet.0 10.10.10.0/24 exact detail
commit

```

retain preserves forwarding state when RPD is unavailable;

no-readvertise prevents redistribution into dynamic routing protocols.

Task 4 - DNS and time zone

Objective

Provide consistent name resolution and administrative time display.

Requirements

14. Use S1 as the DNS server through **mgmt_junos**.
15. Use domain **sys101.lab**.
16. Set **Europe/Amsterdam** as the time zone.
17. Resolve and reach **s1.sys101.lab** through **mgmt_junos**.

Solution

```

configure
set system name-server 172.31.1.100 routing-instance mgmt_junos
set system domain-name sys101.lab
set system time-zone Europe/Amsterdam
commit
run ping s1.sys101.lab routing-instance mgmt_junos count 3

```

The expected DNS answer is **172.31.1.100**. If an older address is returned, rebuild S1 without cache and redeploy it.

Task 5 - Authenticated NTP

Objective

Synchronize every router with S1 using authenticated NTP.

Requirements

18. Configure the node-specific loopback address shown in the addressing table.

19. Use S1 as the operational NTP server through **mgmt_junos**.
20. Use MD5 key ID 1 and key value **workbook**.
21. Trust key ID 1 and associate it with S1.
22. Demonstrate an authenticated, reachable NTP association and synchronized status.

Solution

On each router, replace **<LOOPBACK>** with the address from the table:

```
configure
set interfaces lo0 unit 0 family inet address <LOOPBACK>
set system ntp authentication-key 1 type md5 value
set system ntp trusted-key 1
set system ntp server 172.31.1.100 key 1 routing-instance mgmt_junos
commit
```

Enter **workbook** when prompted. **boot-server** is not available in Junos

25.4R1.12 and is intentionally omitted. The loopback supplies a usable source identity for this platform combination.

```
show configuration system ntp | display set | except value
show ntp associations
show ntp status
```

Wait for nonzero reach and **sync_ntp**. The association should show **SKEY**.

Task 6 - Configuration archival over SCP

Objective

Archive configuration automatically after every commit.

Requirements

23. Use S1 SCP with **lab / lab123** through **mgmt_junos**.
24. Store files in **/srv/ftp/archive** on S1.
25. Configure transfer on commit.
26. Trigger at least one post-configuration archive.
27. Demonstrate a non-empty archived configuration on S1.

Solution

```
configure
set system archival configuration transfer-on-commit
```

```
set system archival configuration routing-instance mgmt_junos
set system archival configuration archive-sites "scp://lab@172.31.1.100/srv/ftp/archive" password
commit
```

Enter **lab123** when prompted, then make one harmless committed change. Verify:

```
show configuration system archival
show log messages | match transfer

find services/archive -type f -size +0c -ls
```

The tested vJunos FTP client rejects the internal routing-instance option.

SCP is therefore the supported lab method. For manual testing:

```
file copy /config/juniper.conf.gz scp://lab@172.31.1.100/srv/ftp/archive/R1-test.conf.gz routing-instance
mgmt_junos
```

Task 7 - Authorization classes and RADIUS mappings

Objective

Map RADIUS identities to distinct Junos authorization profiles.

Requirements

28. **ops-class** must have clear, network, reset, trace, and view permissions.
29. **noc-class** must have all permissions but deny clear, configure, edit, and start-shell commands.
30. Create local passwordless templates **radius-ops** and **radius-noc** for the two classes.
31. Create local passwordless template **remote** with read-only authorization.
32. Map RADIUS users **ops** and **noc** to their corresponding local templates with **Juniper-Local-User-Name**.
33. Demonstrate permitted and denied actions for each constrained role and read-only authorization for an unmapped RADIUS user.

Solution

```
configure
set system login class ops-class permissions clear
set system login class ops-class permissions network
set system login class ops-class permissions reset
set system login class ops-class permissions trace
set system login class ops-class permissions view
set system login class noc-class permissions all
set system login class noc-class deny-commands "(clear | configure | edit | start shell)"
```

```
set system login user radius-ops class ops-class
set system login user radius-noc class noc-class
set system login user remote class read-only
commit
```

These are passwordless local authorization templates, not duplicate local credentials. S1 returns these FreeRADIUS attributes:

```
ops Cleartext-Password := "ops123"
  Juniper-Local-User-Name := "radius-ops"

noc Cleartext-Password := "noc123"
  Juniper-Local-User-Name := "radius-noc"

readonly Cleartext-Password := "readonly123"
```

The unmapped **readonly** identity inherits the special **remote** template. Test both permitted and denied commands after Task 8 activates RADIUS.

Task 8 - RADIUS authentication with local fallback

Objective

Use centralized authentication first while maintaining local recovery access.

Requirements

34. Authentication order must be RADIUS followed by local password.
35. Use S1 through **mgmt_junos** with shared secret **workbook**.
36. Set retry to 1 and timeout to 2 seconds.
37. Preserve local break-glass user **lab / lab123**.
38. Prove RADIUS authentication while S1 is available.
39. Prove local fallback while S1 RADIUS is unavailable without causing administrative lockout.

Solution

Keep the current session open:

```
configure
set system authentication-order radius
set system authentication-order password
set system radius-server 172.31.1.100 secret
set system radius-server 172.31.1.100 routing-instance mgmt_junos
set system radius-server 172.31.1.100 retry 1
```

```
set system radius-server 172.31.1.100 timeout 2
commit confirmed 10
```

Enter **workbook**, then prove login as **ops**, **noc**, and **readonly** from a second terminal. Confirm the commit only after successful tests.

To prove local fallback, temporarily stop RADIUS without removing S1:

```
docker exec S1 supervisorctl stop radius
ssh lab@R3
docker exec S1 supervisorctl start radius
```

lab / lab123 must work while RADIUS is stopped. This proves password fallback; a normal local login while RADIUS is reachable does not prove which method was used. Verify RADIUS traffic/logs on S1 when needed.

Final verification

```
bash scripts/verify.sh
```

Required final line:

```
RESULT: PASS
```

Rollback and reset

```
make destroy
make deploy
```

enforce-startup-config: true reloads the supplied bootstrap baseline on a clean deployment; Containerlab injects the management plane again.